University of Central Punjab

Faculty of Information Technology & Computer Sciences

Data Structures, Spring 2026

Assignment 4

Total Marks: 50	**Deadline: 18 - 06 - 2026**	**Section: D5**	
Name:		**Reg. #:**	
CLO	**CLO Statement**	**Taxonomy Level**	**PLO**
3	Analyze algorithms to understand their structure and functionality.	C5	**3**

Instructions: (5 marks will be deducted for not following the instructions)

- You have to submit Assignment 4 via Portal. Do individually.
- Quiz will be taken on the basis of assignment.
- In case of plagiarism, ***straight ZERO*** will be awarded.
- Attempt all questions in sequence. **Attach this title page as a front page of assignment**.
- Assignments will not be accepted on email and late submissions of assignments are strictly prohibited.
- Attach output screen shot of each question.
- Submit your code with screenshots of the result running over terminal

Answers Clarity: Justify your answer where needed.

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Question 1: A country's emergency communication system consists of thousands of communication towers. The central command tower is the root. Each tower can communicate with at most two subordinate towers. Recently, several communication failures occurred. The government wants software that can analyse the network and identify vulnerabilities. **[20 marks]**

Tasks

Develop a program that:

1. Models the communication network.
2. Identifies towers whose failure would disconnect an entire communication region.
3. Displays every possible communication route from headquarters to local towers.
4. Determines the longest communication chain.
5. Finds the level containing the maximum number of towers.
6. Identifies isolated communication zones after a tower failure.

Sample Output:

```
Vulnerable Towers:
A B C E

Communication Routes:
A B D
A B E H
A B E I
A C F
A C G

Longest Communication Chain:
A B E H

Level with Maximum Towers = 2 (4 towers)

Tower Failed: E

Isolated Communication Region:
E H I
```

Question 2: An AI system stores disease diagnosis paths in a tree. **[10 marks]**

```
                  Diagnosis
                 /          \
          Infection       Deficiency
         /         \      /         \
    Viral     Bacterial Iron     Vitamin
```

The system can analyse diseases using different traversal strategies.

Tasks

Without executing code:

1. Determine which diseases are visited using Preorder.
2. Determine which diseases are visited using Inorder.
3. Determine which diseases are visited using Postorder.
4. Determine which traversal should be used if the diagnosis engine must evaluate parent decisions before child decisions.
5. Determine which traversal should be used if all child diagnoses must be completed before a final decision.

Question 3: A national digital library stores book IDs in an **AVL Tree** or **Red-Black Tree**. As thousands of new books are added every day, the system must automatically maintain balance to ensure efficient search, insertion, and retrieval operations. **[20 marks]**

The following book IDs arrive in the given order:

```
50, 30, 20, 40, 35, 70, 80, 90, 85, 10, 5, 15, 65, 75, 72
```

Tasks

1. Insert the book IDs one by one into an **AVL Tree** or a **Red-Black Tree**.
2. After each insertion:
   - Draw the updated tree.
   - For an AVL Tree:
     - Calculate the balance factor of each affected node.
     - Identify any imbalance.
   - For a Red-Black Tree:
     - Assign appropriate node colours.
     - Identify any violations of Red-Black Tree properties.
3. Whenever an imbalance or violation occurs:
   - For an AVL Tree:
     - Determine whether the case is **LL, RR, LR, or RL**.
     - Perform the required rotation(s).
   - For a Red-Black Tree:
     - Perform the necessary **recolouring and/or rotation(s)** to restore the Red-Black Tree properties.
4. Draw the final balanced AVL Tree or Red-Black Tree.
5. Record:
   - The total number of rotations performed.
   - The total number of recolouring operations (for Red-Black Tree only).